Yu Fang

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Education

University of North Carolina at Chapel Hill — **Fall 2023 - Present**
Ph.D. in Computer Science — *Chapel Hill, NC, US*

ShanghaiTech University — **Fall 2020 - Spring 2023**
M.S. in Computer Science — *Shanghai, China*
IDEA Lab | Advisors: Dinggang Shen, Zhiming Cui

Donghua University — **Fall 2016 - Spring 2020**
B.E. in Software Engineering — *Shanghai, China*

Research Interests

I am broadly interested in **3D computer vision** and **embodied AI**, focusing on bridging the gap between agents and humans, as well as the virtual and physical worlds. My research includes developing spatial intelligence, embodied intelligence, and generative AI for content creation and editing. Currently, I am working on reducing sim-to-real gaps in embodied AI to scale robot learning.

Research Experience

Scaling Robot Learning with Real-to-Sim-to-Real Video Synthesis — **July 2024 - Present**
University of North Carolina at Chapel Hill

- **Motivation**: Large-scale robot data is essential for advancing robot learning. However, collecting real-world data is costly, and simulated data often suffers from the sim-to-real gap. To address these limitations, we aim to develop an end-to-end framework for generating realistic robot videos, leveraging large-scale real-world datasets (DROID and BridgeData V2).
- **Method**: We inpaint the original observations to serve as backgrounds and replay real-world robot trajectories to execute new tasks in the simulation environment (Isaac Sim), ultimately synthesizing novel demonstrations. Using these demonstrations, we fine-tune VLA models (e.g., Octo, OpenVLA) and validate their performance on both simulated and real-world robots.

3D Digital Dentistry — **Jun 2021 - May 2023**
IDEA Lab, ShanghaiTech University

- Develop **implicit representation** model for generating high-quality tooth model with **fine-grained surface details** only from a single modality, e.g., CBCT images or panoramic X-ray images.
- Explore **neural rendering** techniques for decreasing the number of 2D projections for 3D CBCT image reconstruction, to lower x-ray radiation dose while maintaining high reconstruction quality.
- Develop **digital systems** for fusing 3D models from different modalities to improve the efficiency of measurement and diagnosis.

Selected Publications

Geometry-Aware Attenuation Learning for Sparse-View CBCT Reconstruction
Zhentao Liu, **Yu Fang**, Changjian Li, Han Wu, Yuan Liu, Dinggang Shen, Zhiming Cui.
IEEE Transactions on Medical Imaging (TMI), 2024.

SNAF: Sparse-view CBCT Reconstruction with Neural Attenuation Fields
Yu Fang, Lanzhuju Mei, Changjian Li, Yuan Liu, Wenping Wang, Zhiming Cui, Dinggang Shen.
Arxiv Preprint arXiv:2211.17048.

Curvature-enhanced Implicit Function Network for High-quality Tooth Model Generation from CBCT Images
Yu Fang, Zhiming Cui, Lei Ma, Lanzhuju Mei, Bojun Zhang, Yue Zhao, Zhihao Jiang, Yiqiang Zhan, Yongsheng Pan, Min Zhu, Dinggang Shen.
International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), 2022.

A Fully Automatic AI System for Tooth and Alveolar Bone Segmentation from Cone-beam CT Images
Zhiming Cui*,**Yu Fang***, Lanzhuju Mei*, Bojun Zhang*, Bo Yu, Jiameng Liu, Caiwen Jiang, Yuhang Sun, Lei Ma, Jiawei Huang, Yang Liu, Yue Zhao, Chunfeng Lian, Zhongxiang Ding, Min Zhu, Dinggang Shen.
Nature Communications, 2022.

Services

Teaching Assistant	COMP775-001: Image Processing and Analysis [Fall 2024 @ UNC] CS240: Algorithm Design [Spring 2022 @ ShanghaiTech] BME2113: Algorithm Design and Analysis [Fall 2022 @ ShanghaiTech]
Conference Reviewer	ICLR 2025, MICCAI 2025, MICCAI 2024, SIGGRAPH 2024 (Poster), SIGGRAPH 2023 (Poster)
Journal Reviewer	Scientific Data

Skills

Programming Languages	Python, C/C++, Java, SQL, HTML/CSS/JavaScript
Research Tools	PyTorch, Isaac Sim, VR/AR (Meta Quest 3), Blender, Unity, MeshLab, ITK-SNAP, 3D Slicer, Git, NAS, Notion, LaTeX
Communication	English (fluent), Chinese (native)
Hobbies	Photography, Movies, Music, Games